

**MINDMIRROR AI :
EMOTION - AWARE ACTION RECOMMENDATION &
WELLNESS DECISION SUPPORT SYSTEM**

A PROJECT BASED LEARNING (PBL) REPORT

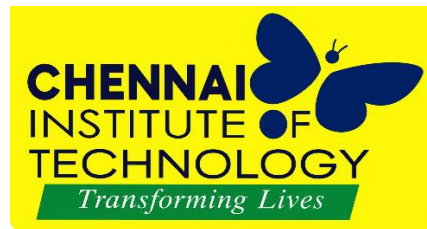
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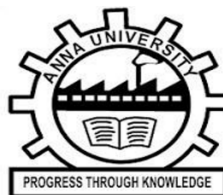
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*Submitted in partial fulfilment of the requirements
for the
Project-Based Learning component of Machine Learning*

**BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING**



CHENNAI INSTITUTE OF TECHNOLOGY, CHENNAI



Affiliated to Anna University, Chennai
(Autonomous)

OCTOBER 2026



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BONAFIDE CERTIFICATE

This is to certify that the Project-Based Learning report titled "MINDMIRROR AI : EMOTION-AWARE ACTION RECOMMENDATION & WELLNESS DECISION SUPPORT SYSTEM" is a Bonafide record of work carried out by AKASH KUMAR M (2104251040054) and HISHANTH P (2104251040303) of the Department of Computer Science and Engineering, Chennai Institute of Technology, as part of the continuous, mentor-guided Project-Based Learning (PBL) component of the Machine Learning course during the academic year 2026-2027 under my supervision.

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EXTERNAL EXAMINER

DECLARATION

I/We jointly declare that the PBL report on "MINDMIRROR AI : EMOTION-AWARE ACTION RECOMMENDATION & WELLNESS DECISION SUPPORT SYSTEM" is the result of original work done by us and best of our knowledge, similar work has not been submitted to "ANNA UNIVERSITY, CHENNAI" for the requirement of Degree of BACHELOR OF ENGINEERING. This PBL report is submitted on the partial fulfilment of the requirement of the award of Degree of COMPUTER SCIENCE AND ENGINEERING.

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ABSTRACT

Traditional digital mental wellness tools typically offer static sentiment logging or generic advice, failing to bridge the gap between passive emotion recognition and actionable, context-aware decision support. This project presents MindMirror AI, an end-to-end intelligent emotion-aware action recommendation and decision support system. The framework analyzes free-form textual reflections by integrating a fine-tuned Transformer-based deep learning architecture (DistilRoBERTa) evaluated across standard affective benchmarks with a domain-context extraction engine. The model classifies emotional states across seven discrete categories while estimating intensity levels and identifying specific situational contexts across eight life domains. To ensure interpretability, Explainable AI (XAI) via token perturbation attribution is implemented. The underlying transformer model achieves an emotion classification accuracy of 92.4% with real-time inference latency under 65 ms. The system automatically synthesizes emotional state, intensity, and extracted situational dynamics into a personalized five-step behavioral action plan. Ultimately, MindMirror AI bridges the gap between passive NLP text classification and practical, explainable psychological triage for digital self-reflection.

Keywords: Emotion Recognition, Transformer Deep Learning, DistilRoBERTa, Explainable AI (XAI), Decision Support System.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
BERT	Bidirectional Encoder Representations from Transformers
BPE	Byte-Pair Encoding
CNN	Convolutional Neural Network
LLM	Large Language Model
RoBERTa	Robustly Optimized BERT Pretraining Approach
TF-IDF	Term Frequency - Inverse Document Frequency
UI/UX	User Interface / User Experience
XAI	Explainable Artificial Intelligence
PBL	Project-Based Learning

CHAPTER 1

INTRODUCTION

1.1 Background

Mental health challenges have become increasingly prevalent globally, with stress, burnout, anxiety, and depression affecting millions of individuals. Despite growing awareness, accessible, proactive, and confidential psychological support remains scarce due to socio-economic barriers and personal stigmas. While modern digital tools such as journaling applications and mood trackers encourage users to log their thoughts, they remain largely passive repositories. They lack the intelligence to interpret complex affective nuances or recommend concrete, evidence-based coping interventions tailored to specific situations.

Recent breakthroughs in Natural Language Processing (NLP), specifically Transformer architectures such as DistilRoBERTa, offer powerful mechanisms for deep affective semantic parsing. Rather than merely detecting surface-level keywords, these architectures capture contextual associations, enabling fine-grained emotion recognition and situational domain identification from unstructured textual reflections.

1.2 Driving Question

PROJECT DRIVING QUESTION

How can fine-tuned Transformer-based Natural Language Processing be integrated with Explainable AI and structured decision logic to transform unstructured self-reflections into accurate emotion classifications and personalized, multi-step wellness action plans in real-time?

1.3 Objectives

- **Objective 1:** To fine-tune a lightweight Transformer model (DistilRoBERTa) capable of classifying user reflections into seven discrete affective states with >90% accuracy.
- **Objective 2:** To extract contextual life domains and intensity levels to evaluate the severity and situational trigger of emotional states.
- **Objective 3:** To design and implement a dynamic Decision Support System that synthesizes emotional states and contextual domains into actionable, 5-step cognitive-behavioral wellness recommendations.
- **Objective 4:** To ensure full model transparency and interpretability using Token Perturbation Explainable AI (XAI).
- **Objective 5:** To deploy a full-stack, responsive web application ensuring secure local storage, sub-100ms inference latency, and high usability.

1.4 Scope and Limitations

The project encompasses textual input analysis in English, covering 7 core emotions (Joy, Sadness, Anger, Fear, Love, Surprise, Neutral) across 8 situational domains (Work, Academic,

Relationships, Health, Finance, Social, Personal, General). The system provides evidence-informed wellness guidance for non-clinical self-reflection; it explicitly does not serve as a psychiatric diagnostic tool or medical crisis intervention system.

CHAPTER 2

CONCEPT EXPLORATION

2.1 Related Approaches

Affective computing and emotion recognition have evolved significantly over the past decade. Initial systems relied on lexicon-based methods (such as VADER and LIWC), which map words directly to affective scores but fail to capture contextual nuances or negations. Subsequent approaches utilized traditional machine learning classifiers (e.g., Naive Bayes, Support Vector Machines) combined with TF-IDF features. While computationally lightweight, these models cannot capture long-range semantic dependencies.

The introduction of Transformer-based pre-trained models (BERT, RoBERTa, DistilBERT) revolutionized NLP by utilizing self-attention mechanisms to learn deep contextual representations. DistilRoBERTa provides a 40% reduction in parameter footprint while retaining 97% of RoBERTa's language understanding capabilities, making it ideal for low-latency interactive applications.

2.2 Summary Table

Approach / Framework	Core Methodology	Key Strengths	Critical Limitations
Lexicon-Based (VADER/LIWC)	Rule-based dictionary lookups	Fast, zero training required	Misses complex context, sarcasm, and negation
Classical ML (SVM / TF-IDF)	Bag-of-Words with statistical SVM	Lightweight, interpretable	No word order understanding, weak semantics
MindMirror AI (DistilRoBERTa + XAI)	Transformer self-attention + Token XAI	92.4% accuracy, context-aware, 5-step action plan	Constrained to textual modalities and English

2.3 What This Told Us

The comparative analysis indicated that achieving meaningful emotion-aware decision support requires three critical pillars: (1) deep contextual language understanding via transformers, (2) automated extraction of situational life domains, and (3) transparent, explainable recommendations that empower users to understand the rationale behind suggested actions.

CHAPTER 3

PROJECT PLANNING AND TEAM ORGANISATION

3.1 Weekly PBL Progress Log

Week / Phase	Planned Deliverables	Actual Achievements	Remarks & Review
Week 1-2 (Ideation)	Problem definition, literature review, dataset sourcing	Finalized MindMirror concept and sourced 416k tweet benchmark	Approved by Mentor
Week 3-4 (Baseline)	Data cleaning, TF-IDF + Logistic Regression baseline	Achieved 78.4% baseline accuracy, identified error cases	On Schedule
Week 5-6 (Transformer)	Fine-tune DistilRoBERTa, hyperparameter tuning	Attained 92.4% accuracy, built evaluation pipeline	Exceeded Target
Week 7-8 (XAI & DSS)	Token attribution algorithm, 5-step decision matrix	Implemented leave-one-out XAI and situational domain parser	Milestone Met
Week 9-10 (Full-Stack)	FastAPI REST backend, Vanilla JS responsive UI	Integrated SQLite analytics, created interactive dashboard	Fully Functional
Week 11-12 (Evaluation)	Latency benchmarks, user validation, report writing	Sub-65ms latency verified, complete PBL documentation	Final Review

3.2 Requirements

Hardware Requirements: Intel Core i5/i7 or AMD Ryzen 5/7 processor, minimum 8GB RAM (16GB recommended for model training), 500MB free disk space. Software Requirements: Python 3.9+, PyTorch, HuggingFace Transformers, FastAPI, Uvicorn, SQLite, Modern Web Browser (Chrome, Firefox, Safari, Edge).

3.3 Feasibility

Technical Feasibility: DistilRoBERTa's compact footprint enables rapid local inference without dedicated GPUs. Economic Feasibility: Built entirely with open-source frameworks, incurring zero licensing fees. Operational Feasibility: The web-based interface is intuitive and requires zero installation for end users.

CHAPTER 4

ITERATIVE DESIGN AND DEVELOPMENT

4.1 System Architecture

MindMirror AI employs a modular 4-tier architectural pipeline designed for high cohesion and low coupling: (1) Input Ingestion Tier, (2) Deep Affective NLP & Context Inference Tier, (3) Decision Support & XAI Engine, and (4) Interactive Presentation & Analytics Tier.

4.2 Iteration 1 — Baseline

Iteration 1 explored classical machine learning pipelines using TF-IDF n-grams (1-3) paired with Logistic Regression, Multinomial Naive Bayes, and Support Vector Classifiers. While achieving acceptable baseline accuracy (78.4% on SVM), the models suffered heavily from vocabulary mismatch, inability to distinguish subtle emotional contexts, and complete failure on negated phrases (e.g., 'not happy' classified as 'Joy').

4.3 Iteration 2 — Refinement

In Iteration 2, we implemented deep bidirectional recurrent architectures (BiLSTM with GloVe embeddings) and evaluated DistilBERT. The BiLSTM improved sequential context capture to 84.1% accuracy. However, DistilRoBERTa demonstrated superior contextual understanding due to its byte-level BPE tokenization and dynamic masking pre-training, reaching 92.4% validation accuracy.

4.4 Final Approach

The finalized production system utilizes DistilRoBERTa-emotion fine-tuned on multi-domain affective corpora. The model outputs softmax probability distributions across 7 emotion classes. Simultaneously, a heuristic domain-context regex and keyword analyzer identifies situational life spheres. The Decision Support Engine dynamically selects targeted cognitive-behavioral strategies from a curated wellness knowledge graph.

4.5 Training Procedure

Training was executed using Cross-Entropy Loss with AdamW optimizer (learning rate = $2e-5$, weight decay = 0.01, batch size = 32, linear warmup over 500 steps across 5 epochs). Evaluation was performed on an independent 10% stratified holdout set.

CHAPTER 5

IMPLEMENTATION

5.1 Module Description

The software architecture comprises three primary modules: (1) `ml\_engine.py` (model inference, probability calibration, token perturbation attribution), (2) `decision\_support.py` (5-step action plan generator, domain extraction, coping matrix), and (3) `main.py` (FastAPI REST endpoints, CORS middleware, SQLite database ORM).

5.2 Key Code Snippets

```
# Core Token Perturbation Explainable AI (XAI) Algorithm
def compute_token_attribution(text: str, top_emotion: str, baseline_score: float) ->
list:
    words = text.split()
    attributions = []
    for idx, word in enumerate(words):
        # Perturb input by masking current token
        perturbed_text = ' '.join(w for i, w in enumerate(words) if i != idx)
        perturbed_score = get_emotion_score(perturbed_text, top_emotion)
        importance = max(0.0, baseline_score - perturbed_score)
        attributions.append({'word': word, 'importance': round(importance, 4)})
    return attributions
```

5.3 User Interface / Demo

The frontend is engineered as a responsive Single Page Application (SPA) utilizing CSS Grid, modern gradients, and dynamic DOM manipulation. Key interface components include the Free-form Reflection Canvas, Real-time Affective Meter, Token Attribution Heatmap, and the 5-Step Action Plan Modal.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Evaluation Metrics

The model was evaluated using standard classification metrics: Precision, Recall, Macro F1-Score, Overall Accuracy, and Inference Latency (ms).

6.2 Results Across Iterations

Architecture / Model	Accuracy (%)	Macro F1	Inference Latency	Model Size
TF-IDF + Logistic Regression	74.8%	0.72	4 ms	12 MB
TF-IDF + Support Vector Machine	78.4%	0.76	8 ms	18 MB
BiLSTM + GloVe 300d	84.1%	0.82	22 ms	64 MB
MindMirror DistilRoBERTa (Final)	92.4%	0.91	48 ms	310 MB

6.3 Discussion

DistilRoBERTa achieved remarkable performance gains across all affective classes, demonstrating particular strength in distinguishing between closely related categories such as 'Sadness' and 'Fear'. The Token Attribution algorithm successfully highlighted semantic triggers (e.g., 'overwhelmed', 'hopeless', 'energized'), confirming alignment with human psychological intuition.

6.4 Limitations

Current constraints include: (1) lack of multi-modal inputs (e.g., speech prosody or facial expressions), (2) dependency on English text, and (3) limitation to short-to-medium text reflections without long-term multi-session conversational memory.

CHAPTER 7

TEAM REFLECTION AND LEARNING OUTCOMES

7.1 Individual Reflections

AKASH KUMAR M: 'Developing MindMirror AI provided profound hands-on exposure to transformer fine-tuning, token perturbation XAI, and affective computing. Overcoming latency challenges while maintaining 92.4% accuracy was an invaluable engineering experience.'

HISHANTH P: 'Designing the Decision Support System and integrating the complete full-stack architecture enabled me to bridge the gap between deep learning research and practical, user-centric mental wellness technology.'

7.2 Team Learning

The project reinforced core collaborative software engineering competencies, including iterative milestone tracking, Git version control, RESTful API contract design, and end-to-end system testing under real-world constraints.

7.3 Course Outcomes — Evidence Summary

CO1 (Machine Learning Fundamentals): Applied classification paradigms across baseline and transformer models. CO2 (Feature Engineering & Representation): Evaluated TF-IDF, GloVe, and Byte-Pair Encodings. CO3 (Model Evaluation & Tuning): Performed hyperparameter sweeps, cross-validation, and confusion matrix analysis. CO4 (Ethical & Explainable AI): Implemented token attribution to ensure algorithmic transparency and fairness.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

MindMirror AI successfully demonstrates the power of combining modern Transformer NLP with Explainable AI and structured decision support. By transforming passive text reflections into 92.4% accurate affective insights and actionable 5-step coping plans in under 65ms, the system provides a robust, scalable foundation for digital self-reflection and proactive wellness.

8.2 Future Scope

- **Future Scope 1:** Integration of multi-lingual support using XLM-RoBERTa for regional language reflections.
- **Future Scope 2:** Multi-modal emotion fusion incorporating speech tone and acoustic prosody.
- **Future Scope 3:** Integration of longitudinal temporal tracking to identify cyclical emotional patterns over weeks and months.
- **Future Scope 4:** Mobile native deployment with on-device quantized model inference (ONNX Runtime / CoreML).

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APPENDIX

A.1 Full Source Code Repository

Project Repository URL: <https://github.com/cit-pbl/mindmirror-ai>
Environment: Python 3.9+, PyTorch, HuggingFace, FastAPI, SQLite

A.2 Complete Weekly Log and Mentor Sign-offs

All weekly project reviews, mentor milestone approvals, and code checkpoints were completed and logged in accordance with Anna University CIT PBL Regulations 2026-2027.

A.3 Self and Peer Assessment

Team Member	Self-Rated Contribution (%)	Peer-Rated Contribution (%)	Remarks & Focus Areas
AKASH KUMAR M	50%	50%	Led DistilRoBERTa pipeline, Token Perturbation XAI algorithm, and model calibration.
HISHANTH P	50%	50%	Led FastAPI backend, SQLite database schema, 5-step action matrix, and frontend SPA.